

Yang-Baxter equations and the symmetric groups

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Outline

- Yang-Baxter equation
- Connection to symmetric group
- Classification attempts and encountered structure

The Yang-Baxter equation

Let V be a vector space and $R \in \text{End}(V \otimes V)$.

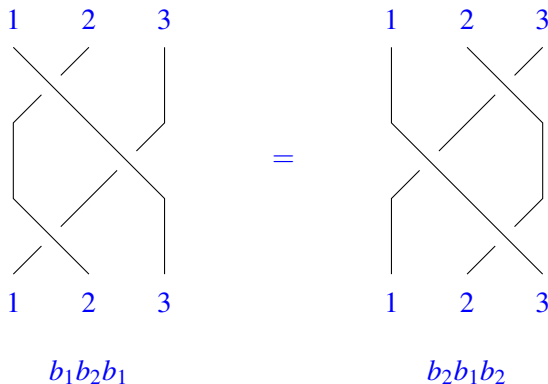
The Yang-Baxter equation on $V \otimes V \otimes V$:

$$R_1 R_2 R_1 = R_2 R_1 R_2,$$

where $R_1 = R \otimes \mathbb{1}$ and $R_2 = \mathbb{1} \otimes R$.

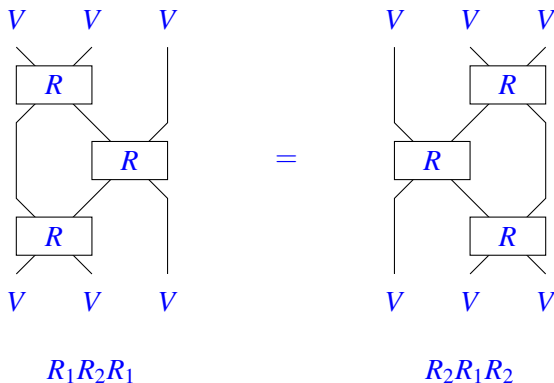
If R is invertible, the Yang-Baxter equation is just the defining relations of B_3 , the braid group on 3 strands.

The braid group defining relations



Presentation: $B_3 = \langle b_1, b_2 | b_1 b_2 b_1 = b_2 b_1 b_2 \rangle$

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Representation on $V^{\otimes 3}$: $b_1 \mapsto R \otimes \mathbb{1}$, $b_2 \mapsto \mathbb{1} \otimes R$.

The braid group defining relations

$$\begin{array}{ccc}
 \cdots i & i+1 & i+2 \cdots \\
 \text{Diagram 1} & = & \text{Diagram 2} \\
 \cdots i & i+1 & i+2 \cdots
 \end{array}$$

$b_i b_{i+1} b_i$
 $\qquad\qquad\qquad$
 $b_{i+1} b_i b_{i+1}$

Presentation:

$$B_n = \langle b_1, \dots, b_{n-1} \mid b_i b_{i+1} b_i = b_{i+1} b_i b_{i+1}, \ b_i b_j = b_j b_i, \ |i-j| \geq 2 \rangle$$

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Representation on $V^{\otimes n}$: $b_i \mapsto \mathbb{1}^{\otimes(i-1)} \otimes R \otimes \mathbb{1}^{\otimes(n-i-1)}$.

Symmetric solutions of YBE

A YBE solution, R , such that $R^2 = \mathbb{1}$ is called *symmetric*. It satisfies the defining relations of the symmetric group

$$S_n = \langle s_1, \dots, s_{n-1} \mid s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, s_i^2 = 1, s_i s_j = s_j s_i, |i-j| \geq 2 \rangle$$

Symmetric solutions define representations $\rho_R^{(n)}$ of S_n on $V^{\otimes n}$

$$\rho_R^{(n)}(s_i) = \mathbb{1}^{\otimes(i-1)} \otimes R \otimes \mathbb{1}^{\otimes(n-i-1)} = R_i.$$

Symmetric solutions of YBE

We define two solutions R, R' to be equivalent, $R \sim R'$, if $\rho_R^{(n)}, \rho_{R'}^{(n)}$ are equivalent as representations for all n .

Lemma

Two symmetric YBE solutions R, R' are equivalent if and only if

$$\mathrm{Tr}_{V^{\otimes n}} (R_1 R_2 \cdots R_{n-1}) = \mathrm{Tr}_{V^{\otimes n}} (R'_1 R'_2 \cdots R'_{n-1}), \quad \forall n \geq 2.$$

Question: How many $t^{(n)}(R)$ are needed to determine R ?

Question: Can all inequivalent R be classified?

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Finite group representation theory

Let G be a finite group.

- For each conjugacy class there is an irreducible representation.

($g, g' \in G$ are conjugate if $\exists h \in G$ such that $g = hg'h^{-1}$.)

Symmetric group conjugacy classes: cycle shape of permutations.

Conjugacy classes of $S_n \leftrightarrow$ partitions of n .

E.g. $(1, 2, 3)(4, 5)$ has cycle shape $[3, 2]$ in S_5 .

- For $g \in G$ and $\rho : G \rightarrow \text{Aut}(V)$ a representation:

Character $\text{ch}_\rho(g) = \text{Tr}(\rho(g))$

Normalised character $\text{nch}_\rho(g) = \frac{\text{Tr}(\rho(g))}{\text{Tr}(\rho(1))}$

Traces in representations are class functions: depend only on conjugacy classes of elements.

- Characters of irreducible representations form an orthonormal basis with respect to

$$\langle \rho, \tilde{\rho} \rangle = \frac{1}{|G|} \sum_{g \in G} \text{ch}_\rho(g) \overline{\text{ch}_{\tilde{\rho}}(g)}$$

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Back to symmetric solutions of YBE

Symmetric solutions of the YBE satisfy

- $\text{ch}_{\rho_R^{(n)}}(1) = \text{Tr}_{V^{\otimes n}}(\mathbb{1}^{\otimes n}) = (\dim V)^n$
- $\text{nch}_{\rho_R^{(m)}}(R_1 R_2 \cdots R_n) = \text{nch}_{\rho_R^{(m+1)}}(R_1 R_2 \cdots R_n), \quad m \geq n + 1$
- $\text{nch}(R_i R_j) = \text{nch}(R_i) \text{nch}(R_j), \quad |i - j| \geq 2$
- For two permutations p, q with disjoint support

$$\text{nch}_{\rho_R^{(n)}}\left(\rho_R^{(n)}(p) \rho_R^{(n)}(q)\right) = \text{nch}_{\rho_R^{(n)}}\left(\rho_R^{(n)}(p)\right) \text{nch}_{\rho_R^{(n)}}\left(\rho_R^{(n)}(q)\right)$$

This property characterises irreducible “representations” of the infinite symmetric group S_∞ .

Infinite symmetric group

Definition S_∞ : Set of bijective maps $\mathbb{N} \rightarrow \mathbb{N}$ that differ from the identity map in only finitely many places.

Infinite group, so definition of irreducibility needs to be refined.

Definition:

A normalised class function χ is irreducible if it cannot be written as a non-trivial convex sum, i.e. there exist no normalised class functions χ_1, χ_2 and $t \in [0, 1]$ such that

$$\chi(-) = t\chi_1(-) + (1-t)\chi_2(-).$$

Infinite symmetric group

Theorem: Thoma

The irreducible normalised class functions of S_∞ are in bijection with pairs of real sequences $\alpha_1, \alpha_2, \dots$ and $\beta_1, \beta_2, \dots$ which satisfy

- ① $\alpha_i \geq 0$ and $\beta_i \geq 0$, for $i \geq 1$.
- ② $\alpha_i \geq \alpha_{i+1}$ and $\beta_i \geq \beta_{i+1}$ for $i \geq 1$.
- ③ $\sum_{i \geq 1} \alpha_i + \sum_{i \geq 1} \beta_i \leq 1$.

The class function evaluated at a group element of cycle shape $\lambda = [\lambda_1, \lambda_2, \dots]$ is

$$\chi(\lambda) = \prod_{j=2}^{\infty} \chi(j)^{m_\lambda(j)}, \quad \chi(j) = \sum_{i \geq 1} \alpha_i^j + (-1)^{j+1} \sum_{i \geq 1} \beta_i^j$$

The normalised traces of R matrices satisfy the same factorisation property as the irreducible normalised class functions of S_∞ .

The normalised traces of symmetric R matrices define irreducible normalised class functions of S_∞ .

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Infinite symmetric group: Class function examples

Examples of Thoma series:

- $\alpha_i = \beta_i = 0, i \geq 1$, the regular representation on $\mathbb{C}[\mathbb{S}_\infty]$.
- $\alpha_1 = 1, \alpha_{i+1} = \beta_i = 0, i \geq 1$, the trivial representation.
- $\beta_1 = 1, \alpha_i = \beta_{i+1} = 0, i \geq 1$, the sign representation.

Question: Which Thoma series correspond to symmetric solutions of the YBE?

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Rational symmetric YBE solutions

Definition

We call a symmetric solution of the YBE rational if its associated Thoma series satisfy

- 1 The series are finite: only finitely many α_i, β_i are non-zero.
- 2 $\alpha_i, \beta_i \in \mathbb{Q}, i \geq 1$.
- 3 $\sum_{i \geq 1} \alpha_i + \beta_i = 1$.

Conjecture

All symmetric YBE solutions on finite-dimensional vector spaces V are rational.

Constructing rational solutions

The set of symmetric solutions of the YBE admits some manipulations. For YBE solutions R, R' on V and W , the following are also solutions.

- 1 $-R$ on V .
- 2 $R \otimes R'$ on $V \otimes W$.
- 3 $\tilde{R} = R \boxplus R'$ on $V \oplus W$, where

$$\begin{aligned}\tilde{R}|_{V \otimes V} &= R, & \tilde{R}|_{W \otimes W} &= R' \\ \tilde{R}|_{V \otimes W} &= \tau, & \tilde{R}|_{W \otimes V} &= \tau,\end{aligned}$$

and τ is the tensor flip $\tau(v \otimes w) = w \otimes v$.

The set of symmetric YBE solutions forms a ring (if we allow for a formal inverse of $\boxplus$).

Constructing rational solutions

Manipulation of YBE solutions at the level of Thoma series:

- ① $-R$: $\alpha_i \leftrightarrow \beta_i$
- ② $R \otimes R'$: $\alpha = \{\alpha_i \alpha'_j, \beta_i \beta'_j\}$, $\beta = \{\alpha_i \beta'_j, \beta_i \alpha'_j\}$
- ③ $\tilde{R} = R \boxplus R'$: Let $d = \dim V$, $d' = \dim W$, then
 $\alpha = \{\alpha_i \frac{d}{d+d'}, \alpha'_i \frac{d'}{d+d'}\}$, $\beta = \{\beta_i \frac{d}{d+d'}, \beta'_i \frac{d'}{d+d'}\}$.

Constructing rational solutions

Every rational Thoma series can be constructed by iterated box summation of trivial and sign representations of various dimensions.

Let

- T_n , n dimensional trivial representation,
- S_n , n dimensional sign representation.

Theorem

Rational symmetric YBE solutions are in bijection with pairs of partitions (A, B) .

- For $A_i \in A$, $\rightarrow T_{A_i}$ summand
- For $B_i \in B$, $\rightarrow S_{B_i}$ summand

The dimension of the underlying vector space is $d = |A| + |B|$. The corresponding Thoma series are $\alpha_i = A_i/d$, $\beta_i = B_i/d$.

If the conjecture is true, this classifies all inequivalent symmetric (finite-dimensional) YBE solutions.

Decomposition into irreducibles

Given a rational symmetric YBE solution on V , how does $V^{\otimes n}$ decompose into irreducible representations of S_n ?

Need more S_n representation theory.

- Given some representation with character ψ , the multiplicity of an irreducible representation with character χ is $\langle \psi, \chi \rangle$.
- The character table of the symmetric group is the transition matrix from power sum basis to the Schur polynomial basis of the ring of symmetric polynomials.

$$s_\lambda = \sum_{\mu} \frac{\chi^\lambda(\mu)}{z_\mu} p_\mu, \quad z_\mu = \prod_{i=1}^{\infty} i^{m_i(\mu)} m_i(\mu)!,$$

where $\chi^\lambda(\mu)$ is the character of the irreducible representation corresponding to the conjugacy class of cycle shape λ evaluated at the conjugacy class of cycle shape μ .

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Symmetric polynomials

- Power sums: $\mathbf{p}_m = \sum_{i=1}^k x_i^m$, $\mathbf{p}_\lambda = \mathbf{p}_{\lambda_1} \mathbf{p}_{\lambda_2} \cdots$, λ a partition.
- Schur polynomials: $\mathbf{s}_\lambda = \det A / \prod_{i < j} (x_i - x_j)$, where

$$A = \begin{bmatrix} x_1^{\lambda_1+k-1} & x_2^{\lambda_1+k-1} & \cdots & x_k^{\lambda_1+k-1} \\ x_1^{\lambda_2+k-2} & x_2^{\lambda_2+k-2} & \cdots & x_k^{\lambda_2+k-2} \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{\lambda_k} & x_2^{\lambda_k} & \cdots & x_k^{\lambda_k} \end{bmatrix}$$

- Products of Schur functions can be decomposed into Schur functions: $\mathbf{s}_\lambda \mathbf{s}_\mu = \sum_{\nu} c_{\lambda, \mu}^{\nu} \mathbf{s}_{\nu}$.
Skew Schur polynomials: $\mathbf{s}_{\lambda/\mu} = \sum_{\nu} c_{\mu, \nu}^{\lambda} \mathbf{s}_{\nu}$.

Decomposition into irreducibles

Given a rational symmetric YBE solution on V , how does $V^{\otimes n}$ decompose into irreducible representations of S_n ?

Theorem

Let (A, B) be the partitions labelling a rational symmetric solution of the YBE on V . The multiplicity of the irreducible representation $I(\lambda)$ of S_n in $V^{\otimes n}$ is

$$[V^{\otimes n} : I(\lambda)] = \sum_{\mu \subset \lambda} s_{\mu}(A) s_{\lambda'/\mu'}(B).$$

Corollary

The multiplicity is zero if and only if the Young diagram of λ contains a rectangle of size $(\ell(A) + 1) \times (\ell(B) + 1)$.

If $B = \emptyset$, $[V^{\otimes n} : I(\lambda)] = s_{\lambda}(A)$.

If $A = \emptyset$, $[V^{\otimes n} : I(\lambda)] = s_{\lambda'}(B)$.

Open problems

- Prove the conjecture.
- Given a rational symmetric YBE solution, identify its equivalence class.
- Generalise setup beyond symmetric group: Hecke algebras?
Braid group?

Thank you for coming!